STAFF REVIEW · H446 141 source models

Separate from the eight-page pupil workbook. This copy is in a public repository and is NOT access controlled. Check technical correctness before use.

Printed question mapping: Booklet Q1 = original online Q1; Booklet Q2 = original online Q2; Booklet Q3 = original online Q3

Do Now — source answers

1. A child key pointing to a nonexistent parent.

2. Binary search: O(log n) rather than O(n).

3. AND.

4. Holding a CPU's working data/ALU results.

I Do — source worked steps

1. Subtract £20 from A — Identify the relevant information and make a prediction.

2. Crash before adding to B — Explain the change using correct technical terms.

3. Rollback both changes — Atomicity means the transfer is all-or-nothing. A DBMS transaction can roll back partial changes. Consistency checks valid constraints; isolation keeps concurrent transfers safe; durability preserves committed transfers.

We Do — guided CHECK

1. Two concurrent requests
Identify the relevant information and make a prediction.
Lock/check last seat
Explain the change using correct technical terms.
Commit one booking
Use an atomic inventory check and reservation update inside an appropriately isolated transaction, with row locking or other concurrency controls. Only one should succeed; second sees no remaining seat.

You Do — source mark schemes

1. A Atomicity [1], C Consistency [1], I Isolation [1], D Durability [1], clear atomic/durable explanations [2].

2. Both read old value [2], overwriting changes [2], locking/transaction isolation prevents conflict [2].

3. Redundancy can reduce downtime [2], RAID is not a backup [2], replication can copy mistakes [2], independent tested backups restore deleted/corrupt data [2].

Exit Ticket — source answer

1. False. Replicated deletions can propagate; an independent backup is still needed.